

EC-Council



Digital Forensics Essentials

MODULE 3

Understanding Hard Disks and File Systems



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Hard Disk Drive

SUBTOPIC 1



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Module Objectives

- 1 Understanding Different Types of Storage Drives and their Characteristics
- 2 Understanding the Logical Structure of a Disk
- 3 Understanding the Booting Process of Windows, Linux, and Mac Operating Systems
- 4 Overview of various File Systems of Windows, Linux, and Mac Operating Systems
- 5 Analyzing File Systems using Autopsy and The Sleuth Kit



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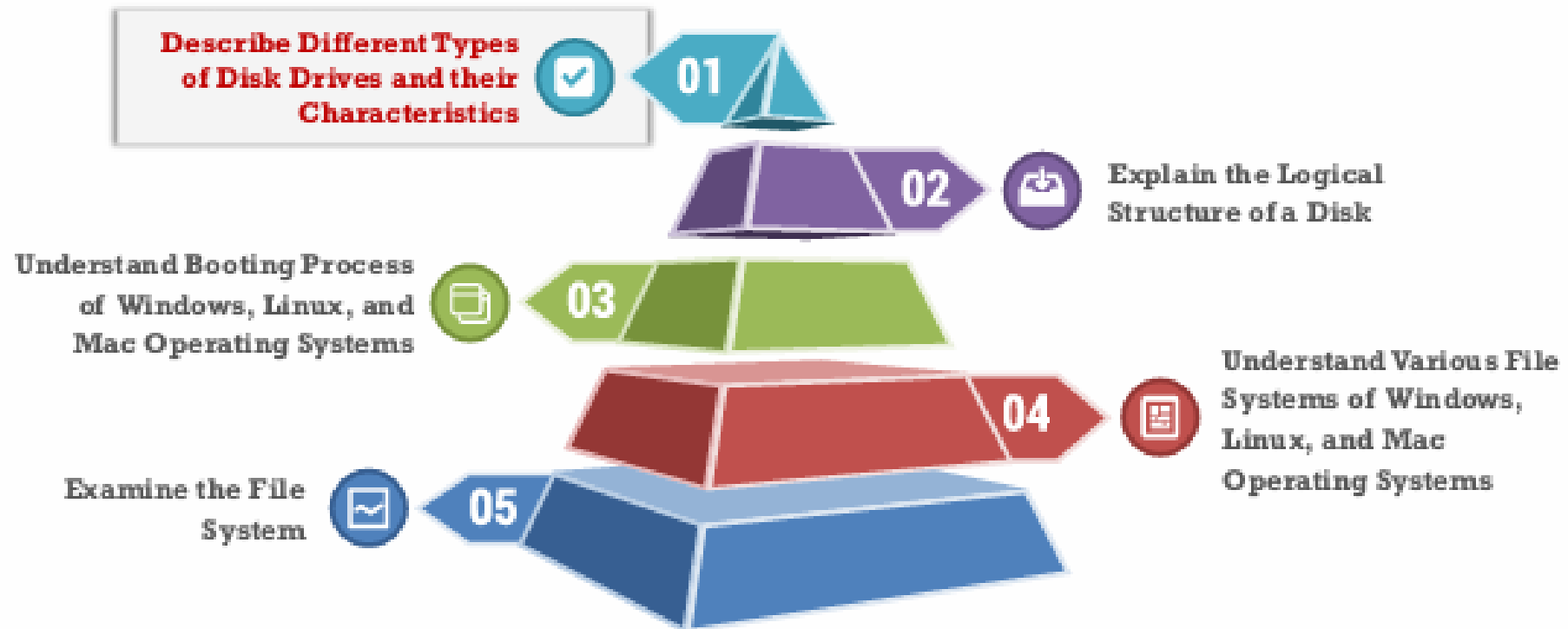
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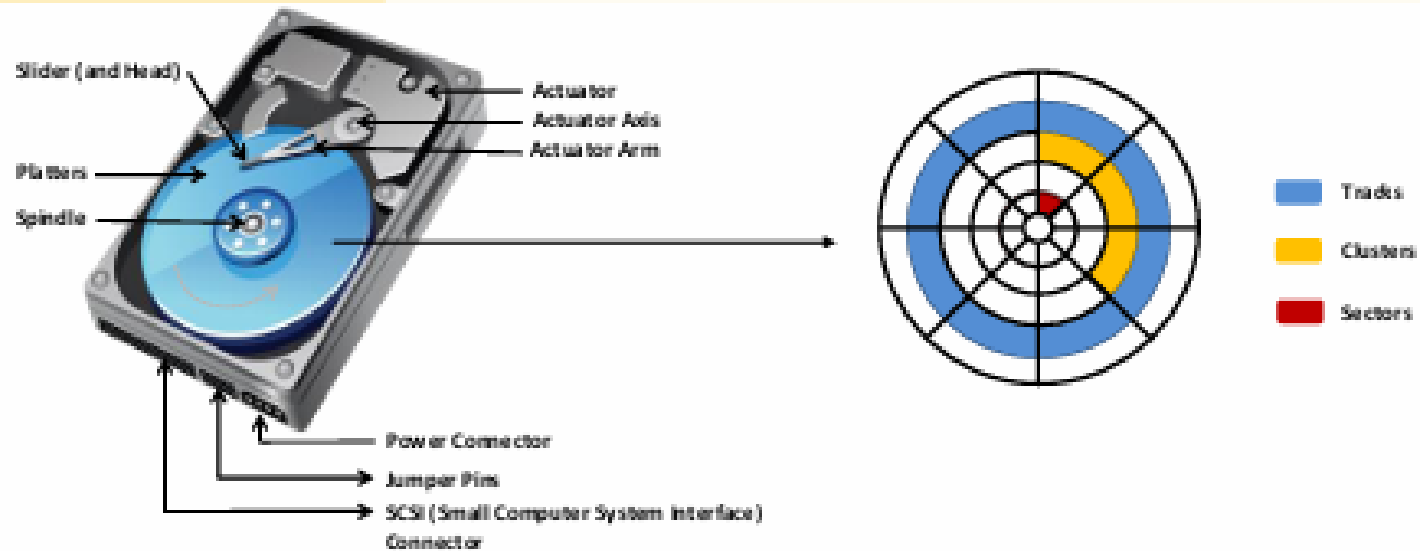
Module Flow



Understanding Hard Disk Drive



- ❑ HDD is a **non-volatile** digital data storage device that **records data magnetically** on a metallic platter
- ❑ The read/write performance of an HDD is directly **proportional to the RPM** (revolutions per minute) of the drive platter



Understanding Hard Disk Drive: Tracks



Tracks are the **concentric circles on platters** where all the information is stored



The drive head can **access** these circular rings in one position at a time



Tracks are numbered for **identification purposes**



Read/write is performed by **rolling headers** from the inner to outermost part of the disk



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Understanding Hard Disk Drive: Track Numbering



Track numbering on a hard disk **begins** at **0** from the outer edge and moves towards the center. The number of tracks on a hard disk depends on the size of the disk



The read/write heads on both surfaces of a platter are tightly packed and locked together on an assembly of head arms



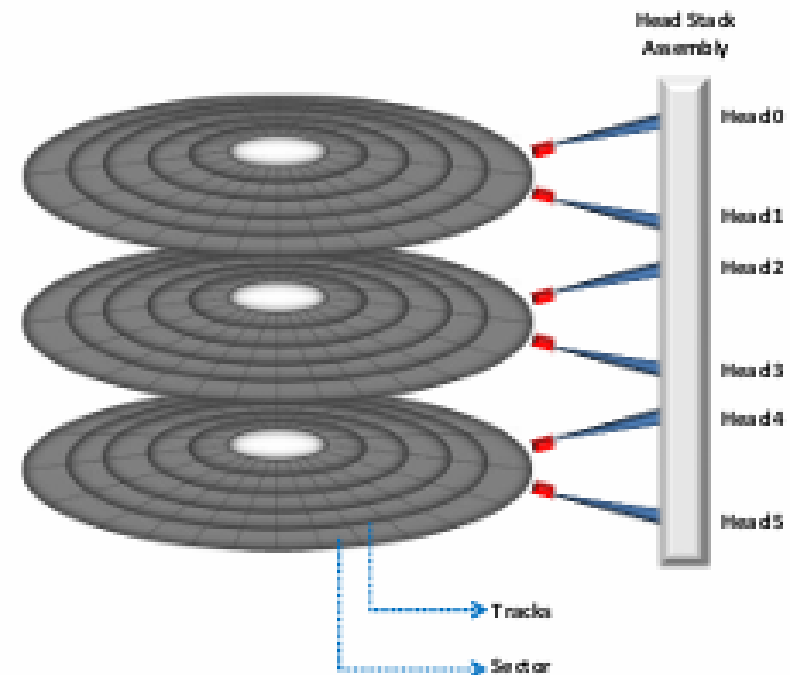
The arms move in and out together to physically locate all heads at the same **track number**



Therefore, a track location is often referred by a cylinder number rather than a track number



A cylinder is a group of all tracks that start at the same head position on the disk





Understanding Hard Disk Drive: Sector

“



A sector is the smallest **physical storage unit** on the disk platter



Each sector holds data of fixed size: **512 bytes for HDDs**, **2048 bytes** for CD-ROMs and DVD-ROMs. Latest HDDs use **4096-byte** (4 KB) sectors.



Each disk sector is labelled using the **factory track-positioning data**



The optimal method of storing a file on a disk is in a **contiguous series**



For example, if the file size is 600 bytes, two 512-bytes sectors are allocated for the file



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Understanding Hard Disk Drive: Sector Addressing

Cylinders, heads, and sectors (CHS) determine the address of the individual sectors on the disk

When a disk is formatted, it is divided into tracks and sectors

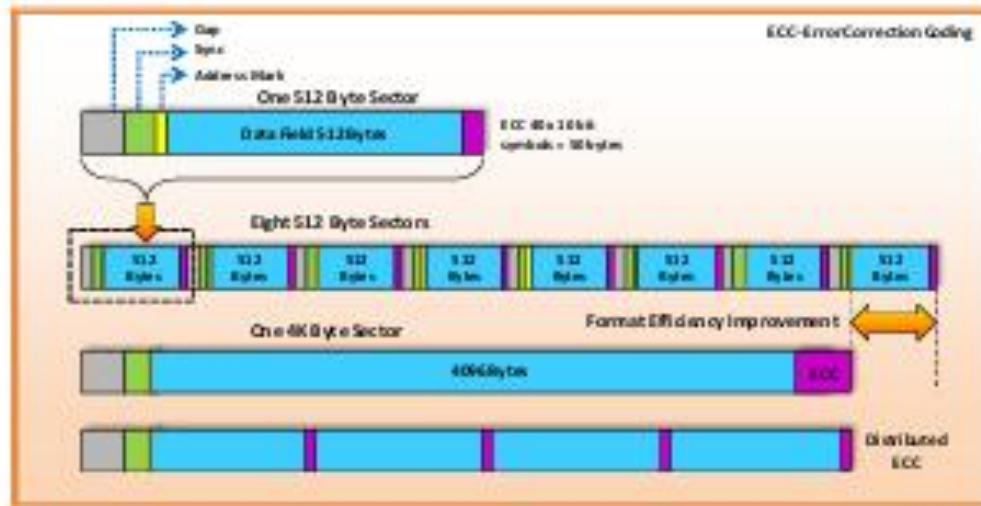
For example, the formatted disk might contain **50 tracks**, each of which is divided into 10 sectors

Track and sector numbers are used by the **OS** and **disk drive** to identify the stored information



Understanding Hard Disk Drive: 4K Sectors

- ❑ New hard drives use **4096-byte** (4 KB or 4K) advanced format sectors
- ❑ Generation-one Advanced Format, also called as 4K sector technology, efficiently uses the **storage surface media** of a disk by merging eight 512-byte sectors into a single sector of 4096 bytes

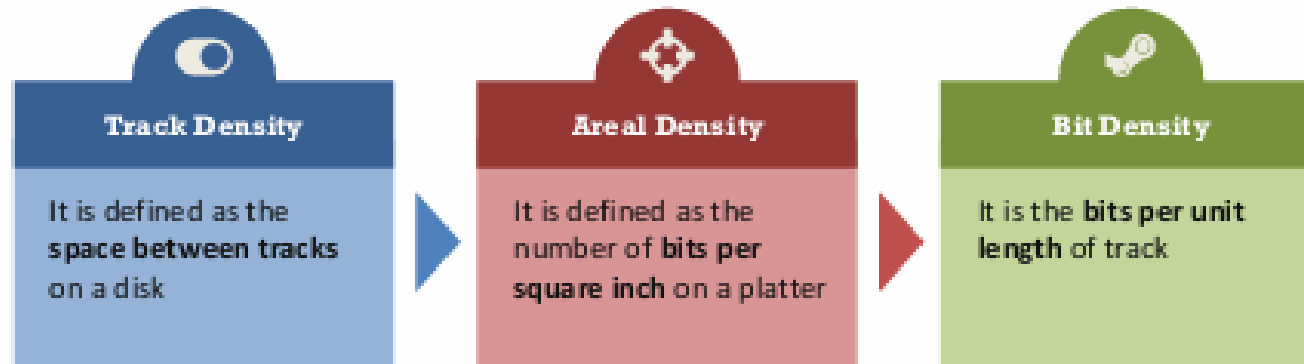


Data Density on a Hard Disk



- ❑ Data is recorded onto a hard disk using a method called **zoned bit recording** (also known as a multiple zone recording)
- ❑ In this technique, tracks are combined together into zones depending on their distance from the **center of the disk**
- ❑ Each zone is assigned a number of sectors per track

Types of data densities on a hard disk:



CHS (Cylinder-Head-Sector) Data Addressing and Disk Capacity Calculation



The **CHS addressing method** addresses each physical block of data on a hard disk by specifying the cylinder (radius), head (platter side), and sector (angular position)

Example of Disk Capacity Calculation:

A disk drive has 16,384 cylinders, 80 heads, and 63 sectors per track. Assume - a sector has 512 bytes. What is the capacity of such a disk?

Answer

Total Size of the Disk = No. of Cylinders * No. of Heads * No. of Sectors per Track * 512 bytes per Sector

Total Size of the Disk = (16,384 cylinders) * (80 heads) * (63 sectors / track) * (512 bytes / sector)

= 42,278,584,320 bytes



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Measuring the Hard Disk Performance



- ❑ Data is stored on the hard disk in the **form of files**
- ❑ When a running program requests a file, the hard disk **recovers the byte content** of the file and sends the bytes to the CPU, one at a time, for further processing

Hard disk performance is measured by these factors:



Data rate: It is a ratio of the **number of bytes per second** that the hard disk sends to the CPU



Seek time: It is the amount of **time required to send the first byte** of the file to the CPU, when it requests the file



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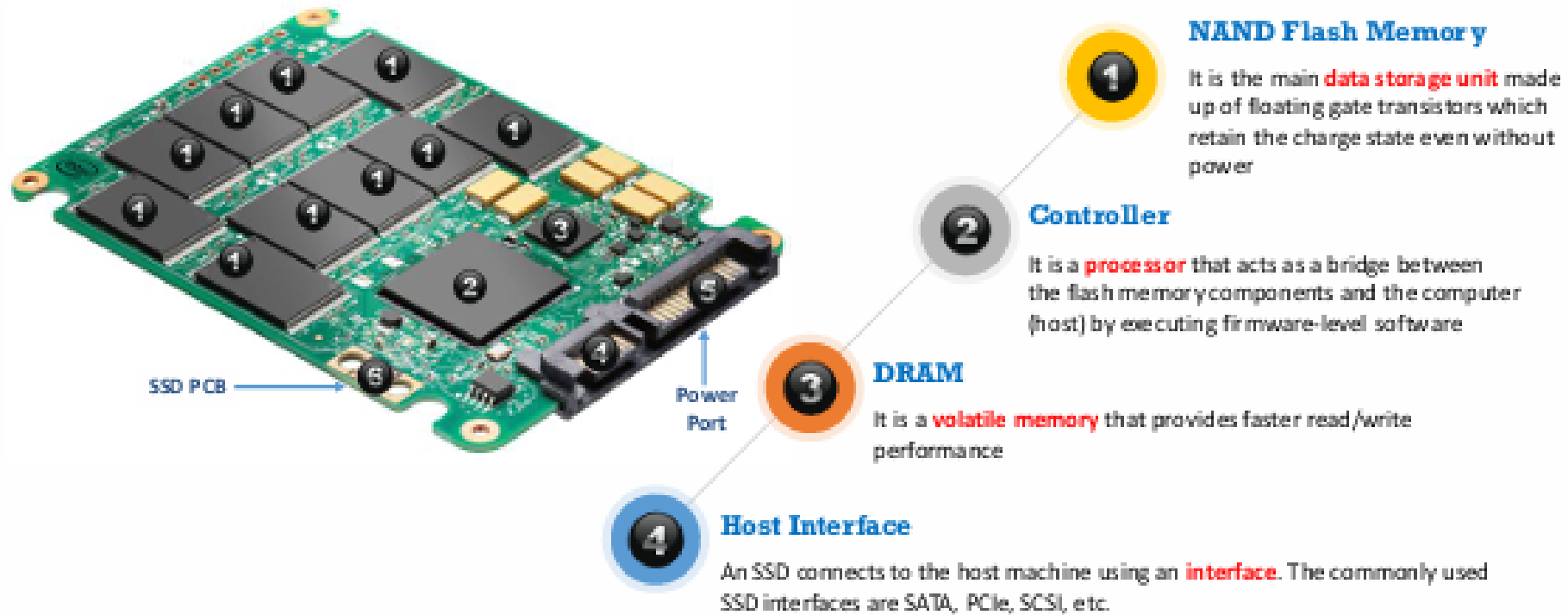


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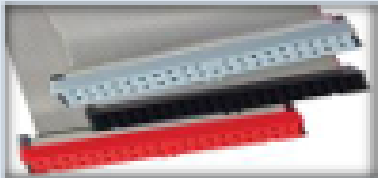
Understanding Solid-State Drive (SSD)

- ❑ SSD is a **non-volatile storage** device that uses **NAND flash memory** chips to store digital data
- ❑ SSDs are **faster** than HDDs as they have **no moving parts**, and the read/write performance depends on data connection of the drive



Disk Interfaces

ATA/PATA (IDE/EIDE)



ATA (**Advanced Technology Attachment**) is the official **ANSI (American National Standards Institute)** name of Integrated Drive Electronics (IDE), a standard interface between a motherboard's data bus and storage disks

Serial ATA/ SATA (AHCI)



It is an **advancement of ATA** and uses serial signaling, unlike IDE's parallel signaling

Serial Attached SCSI



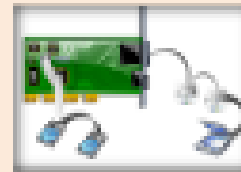
SAS (Serial Attached SCSI) is the successor and an **advanced alternative to parallel SCSI** in enterprise environments

PCIe SSD



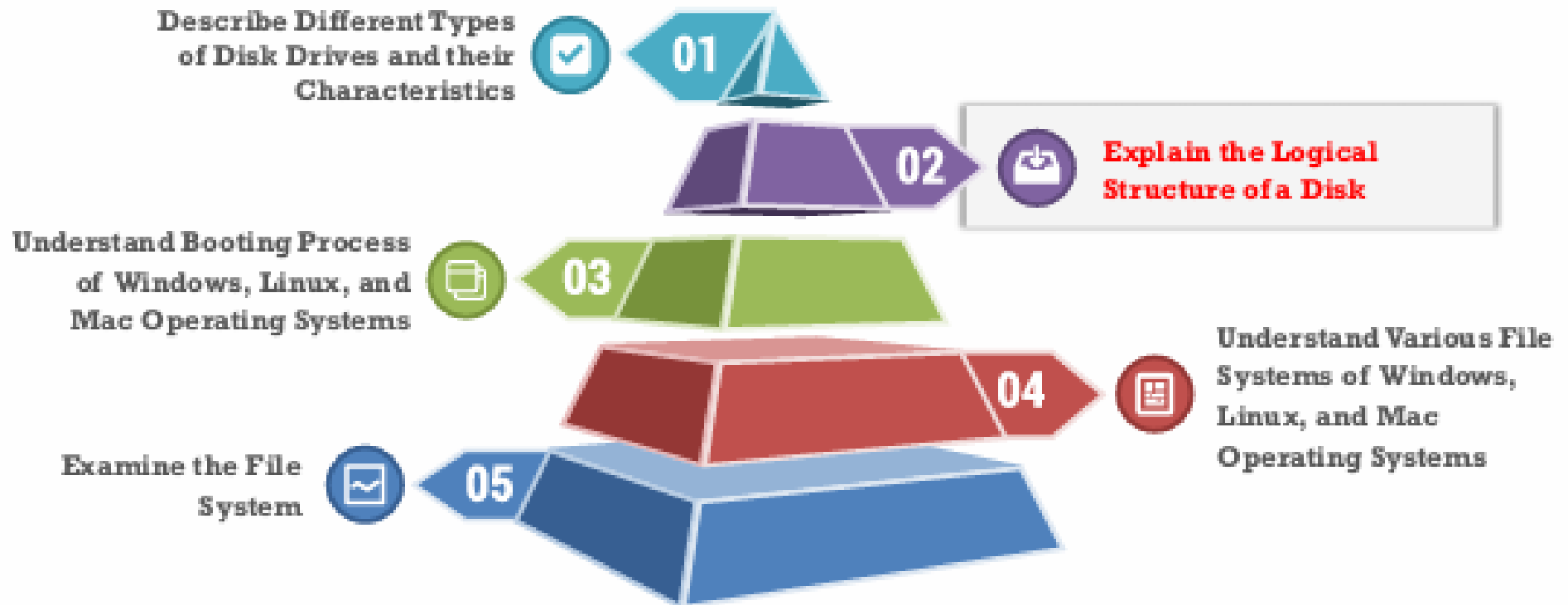
A PCIe (Peripheral Component Interconnect Express) SSD is a **high-speed serial expansion card** that integrates flash directly into the motherboard

SCSI



SCSI (Small Computer System Interface) refers to a set of ANSI standard interfaces based on the parallel bus structure and **designed to connect multiple peripherals** to a computer

Module Flow





Logical Structure of Disks



The logical structure of a hard disk is the **file system and software** utilized to control access to the storage on the disk



A hard disk's logical structure has a significant influence on the **performance, consistency, expandability, and compatibility** of the storage subsystem of the hard disk



Different OSes have different file systems and use various methods of **arranging and controlling access** to data on the hard disk



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Clusters



A cluster is the **smallest logical storage unit** on a hard disk



It is a set of sectors within a disk ranging from cluster number **2 to 32** or more, depending on the formatting scheme in use



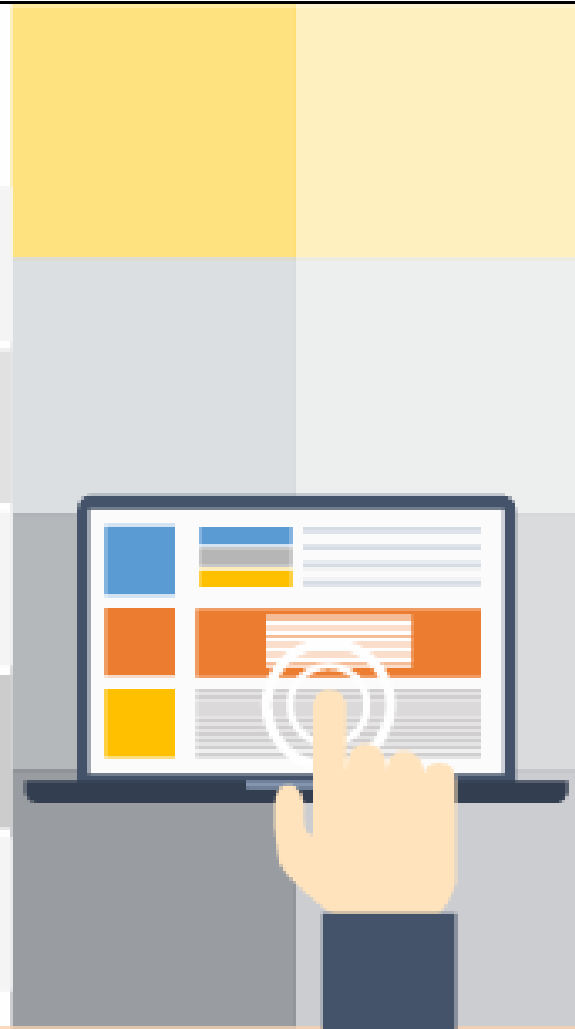
The file system divides the storage on a disk volume into **discreet chunks of data** for efficient disk usage and performance. These chunks are called clusters



The process by which files are allocated to clusters is called allocation; therefore, clusters are also known as allocation units



In the File Allocation Table (FAT) file system, the clusters linked with a file keep **track of file data** in the hard disk's file allocation table



Cluster Size

- Cluster sizing has a significant impact on the **performance of an OS and disk utilization**
- Cluster size can be **altered** for optimum disk storage
- The size of a cluster depends on the **size of the disk partition** and type of file system installed on the partition
- A large cluster size (greater than one sector) has the following effects:
 - Minimizes the **fragmentation** problem
 - Increases the probability of **unused space** in the cluster
 - Reduces the **disk storage** area in which information can be saved
 - Reduces the **unused area** on the disk



Lost Clusters

01

When the OS marks clusters as used but does not **allocate them to any file**, such clusters are known as lost clusters

02

A lost cluster is a FAT file system error that results from the manner in which the FAT file system allocates space and chains files together

03

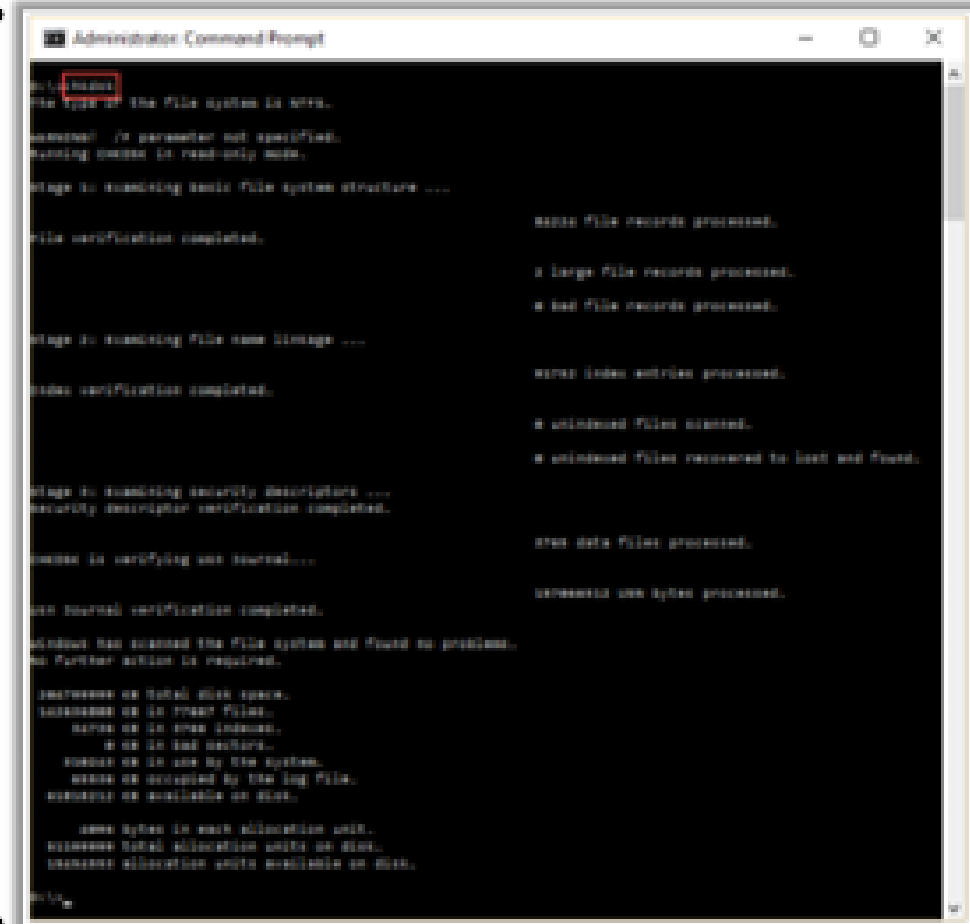
It is mainly the result of a logical structure error and not a physical disk error

04

They usually occur because of **interrupted file activities** caused when, for example, a file is not properly closed; thus, the clusters involved in such activity are never linked correctly to a file

05

CHKDSK is a system tool in Windows that authenticates the **file system reliability of a volume** and repairs **logical file system errors**



```
Administrator Command Prompt
C:\>chkdsk /f
The type of the file system is NTFS.
Warning! /f parameter not specified.
Running CHKDSK in read-only mode.
Stage 1: Examining basic file system structure ...
File verification completed.
Stage 2: Examining file name linkage ...
Index verification completed.
Stage 3: Examining security descriptors ...
Security descriptor verification completed.
Windows is verifying the journal ...
Journal verification completed.
Windows has scanned the file system and found no problems.
No further action is required.
Free space is 100000000 bytes.
Total space is 1000000000 bytes.
Total space on disk is 1000000000 bytes.
Total space in use by the system is 100000000 bytes.
Total space occupied by the log file is 10000000 bytes.
Total space available on disk is 1000000000 bytes.
CHKDSK is now done.
Press any key to continue.
```


Slack Space



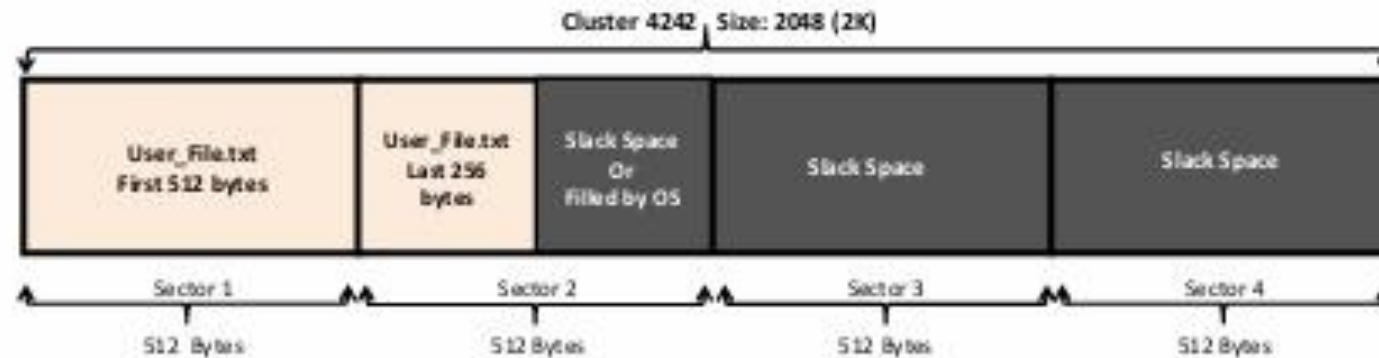
Slack space is the storage area of a disk between the **end of a file** and **the end of a cluster**.



If the file size is less than the cluster size, a full cluster is still assigned to that file. The remaining unused space is called **slack space**.



For example, if the partition size is 4 GB, each cluster will be 32 KB in size. Even if a file requires only 10 KB, the entire 32 KB will be allocated to that file, resulting in 22 KB of slack space.



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Master Boot Record (MBR)

1

A master boot record (MBR) is the **first sector** ("sector zero") of a **data storage device** such as a hard disk

2

The information regarding the files on the disk, their locations and sizes, and other important data is stored in the **MBR** file



3

In practice, MBR almost always refers to the **512-byte boot sector** (or partition sector) of a disk

4

MBR is used for the following:

- **Holding a partition table** which refers to the partitions of a hard disk
- **Bootstrapping** an OS
- Distinctively recognizing individual hard disk media with a **32-bit disk signature**

Structure of a Master Boot Record



The **structure of MBR** consists of three parts:

- ➔ **Master Boot Code or Boot Strap** – It is an executable code and responsible for loading OS into computer memory. It consists of a data structure of **446 bytes**.
- ➔ **Partition Table** – It maintains the data of all the hard disk partitions and consists of a data structure **64 bytes**.
- ➔ **Disk Signature** – It is located at the end of the MBR and contains only **2 bytes** of data. It is required by BIOS during booting.

Address			Description	Size in bytes
Hex	Oct	Dec		
0000	0000	0	Code Area	440 (max. 446)
01B8	0670	440	Disk Signature (Optional)	4
01BC	0674	444	Usually Nulls; 0x0000	2
01BE	0676	446	Table of Primary Partitions (Four 16-byte entries, IBM partition table scheme)	64
01FE	0776	510	55h	MBR Signature; 0xAA55
01FF	0777	511	AAh	
MBR, Total Size: 446 + 64 + 2 =				512

Disk Partitions



- ❑ Disk partitioning is the **creation of logical divisions** on a storage device (HDD/SSD) to allow the user to apply OS-specific logical formatting
- ❑ The disk-partitioning process is the same for both HDDs and SSDs

Primary Partition

- It is a drive that holds the information regarding the **OS, system area**, and other information required for booting
- In MS-DOS and earlier versions of Microsoft Windows systems, the first partition (C:) must be a "primary partition"

Extended Partition

- It is a logical drive that holds the information regarding stored **data and files** in the disk



Globally Unique Identifier (GUID)



The Globally Unique Identifier (GUID) is a 128-bit **unique reference number** used as an identifier in computer software

In general, GUIDs are displayed as **32 hexadecimal digits** with groups separated by hyphens



Common Uses:



In Windows Registry, GUIDs are used to **identify COM (Component Object Model) DLLs (dynamic-link libraries)**



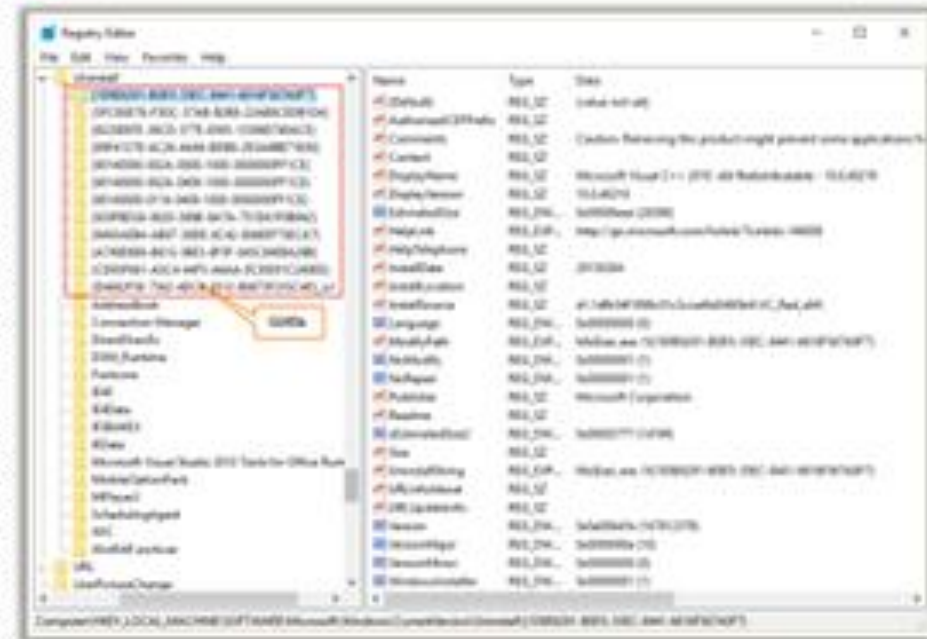
In database tables, GUIDs are used as primary key values



In some instances, a website may assign a GUID to a user's browser to **record and track** the session



Windows assigns a GUID to a username to **identify user accounts**



GUID Partition Table (GPT)

- ❑ **Unified Extensible Firmware Interface** (UEFI) replaces legacy **BIOS firmware** interfaces
- ❑ UEFI is a specification that defines a **software interface** between an OS and platform firmware
- ❑ It uses a partition system known as **GUID Partition Table** (GPT), which replaces the traditional **MBR**

Advantages of GPT disk layout:



Supports up to 128 partitions and uses 64-bit **Logical Block Addresses** (LBAs)

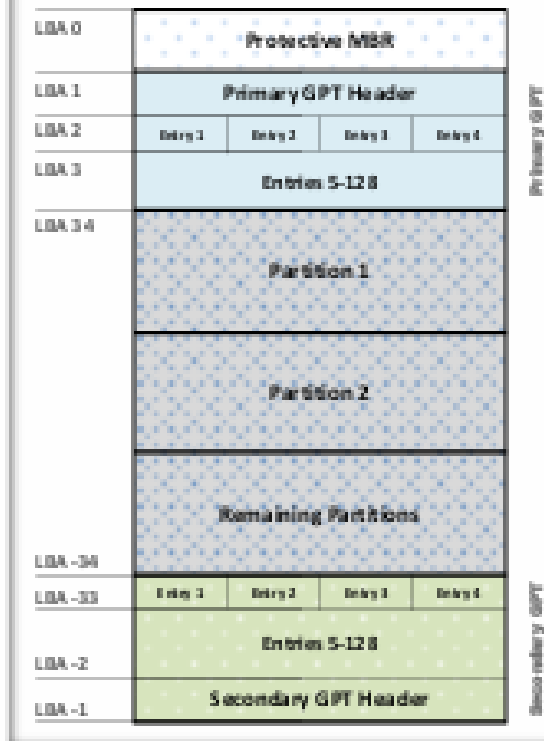


Supports a maximum **partition size** ranging from 2 Terabytes (TiB) to 8 Zebibytes (ZiB)

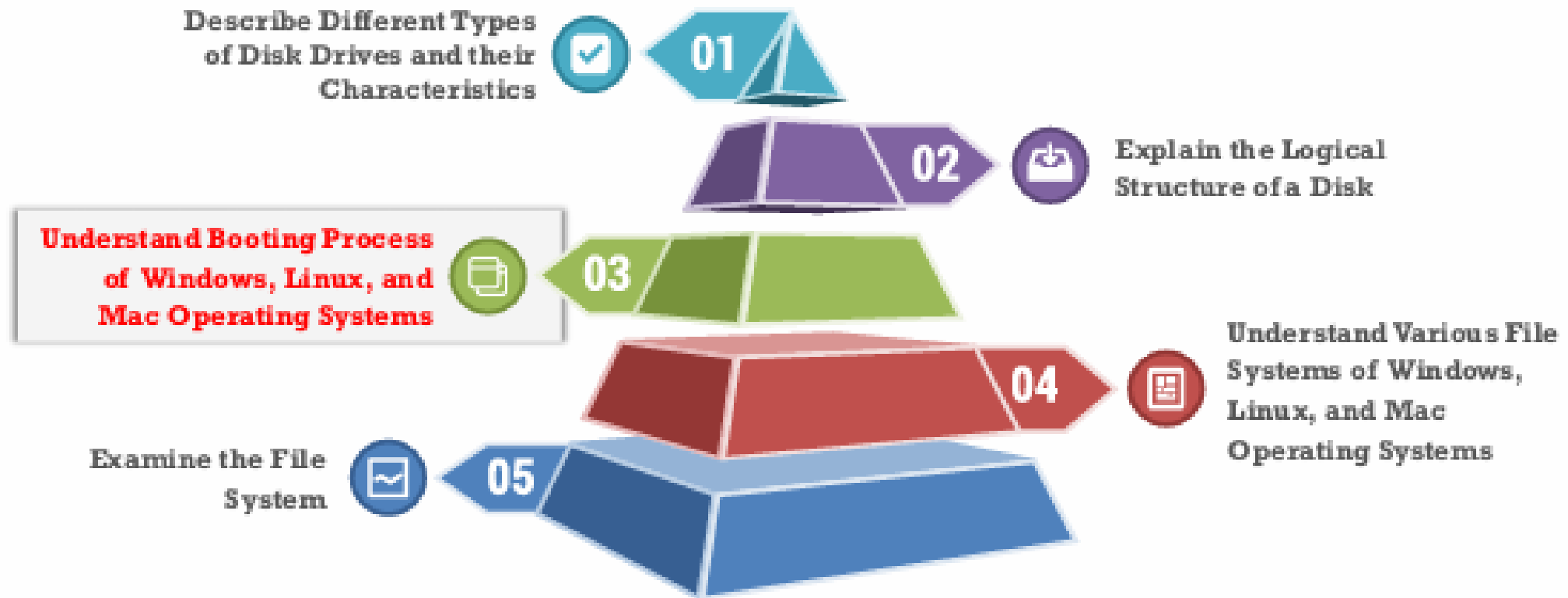


Provides **primary** and **backup partition tables** for redundancy

GUID Partition Table Scheme



Module Flow



What is the Booting Process?

- ❑ Booting refers to the process of **starting or restarting the OS** when the user turns on a computer system
- ❑ It **loads the OS** (stored in the hard disk) to the RAM (working memory)



Types of Booting



Cold boot (Hard boot)

- ✓ It is the process of starting a computer from a powered-down or **off** state



Warm boot (Soft boot)

- ✓ It is the process of restarting a computer that is already **turned on**. A warm boot might occur when the system encounters a **program error** or requires a restart to make certain changes after installing a program, etc.



Understand Booting Process of Windows, Linux, and Mac Operating Systems

SUBTOPIC 2



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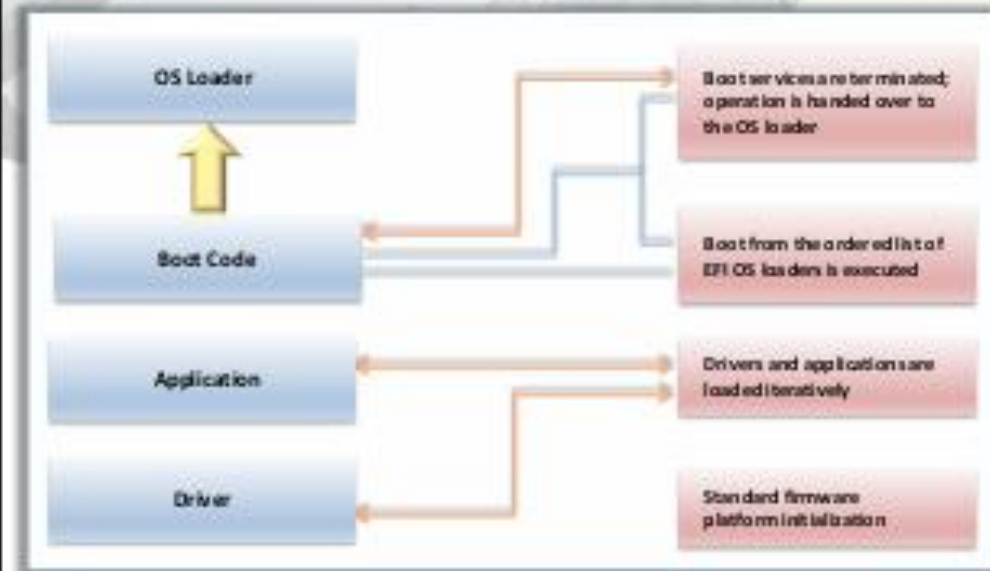
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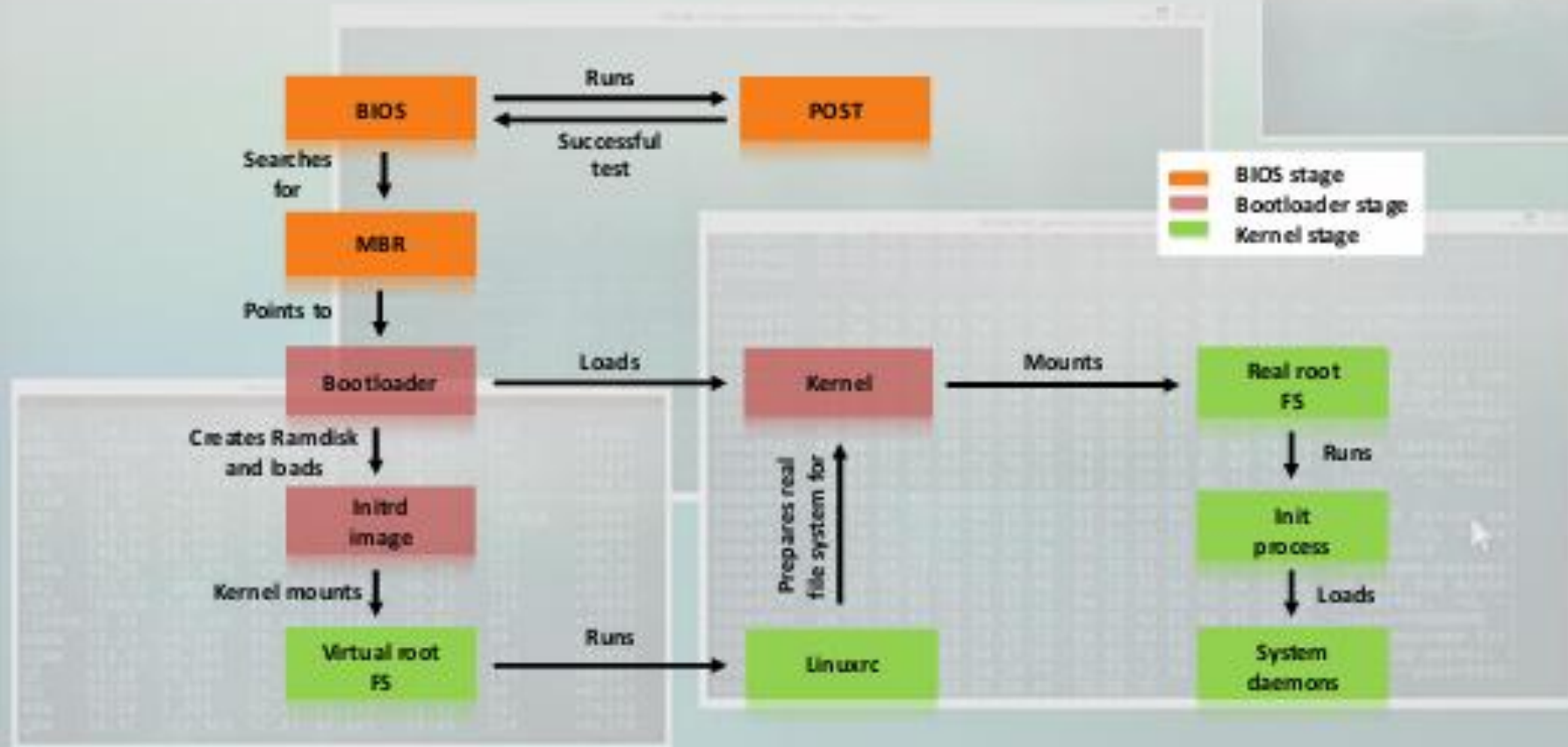
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Macintosh Boot Process

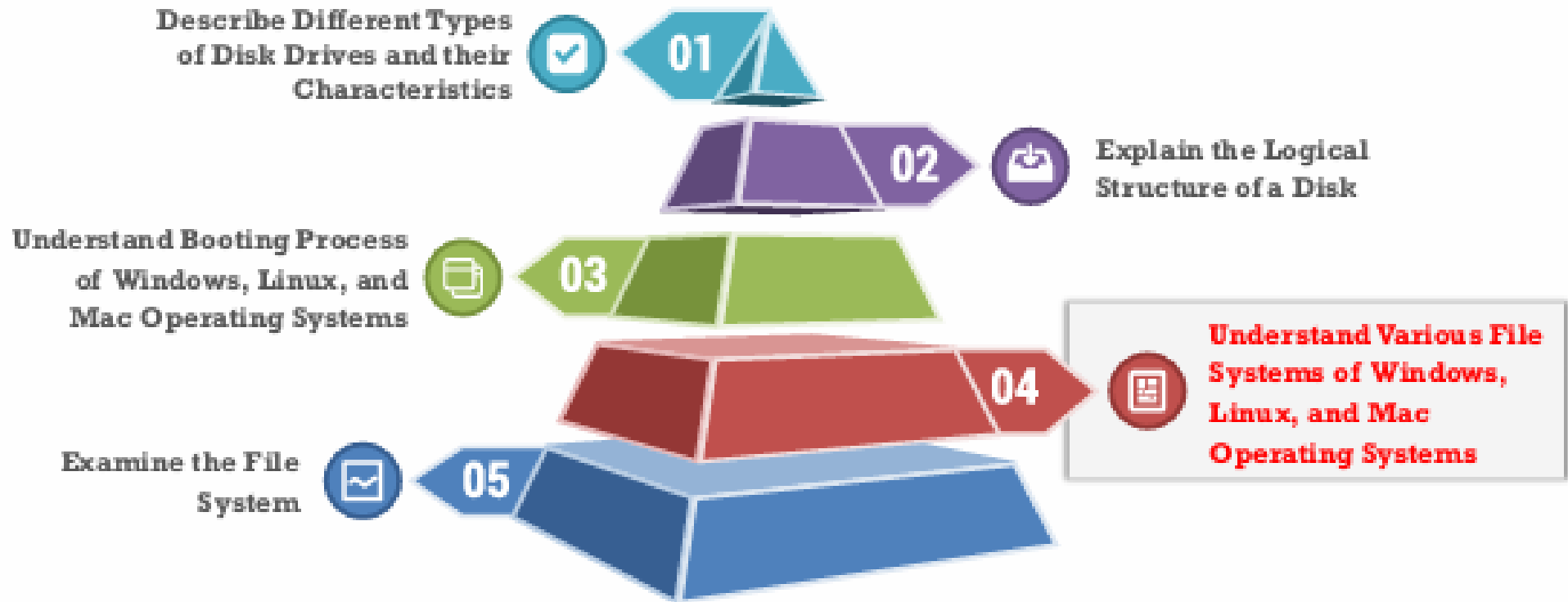


Legend	
	EFI Binaries
	Boot Manager
	Value Added Implementation
	API - Specified
	Upon Encountering an Error

Linux Boot Process



Module Flow

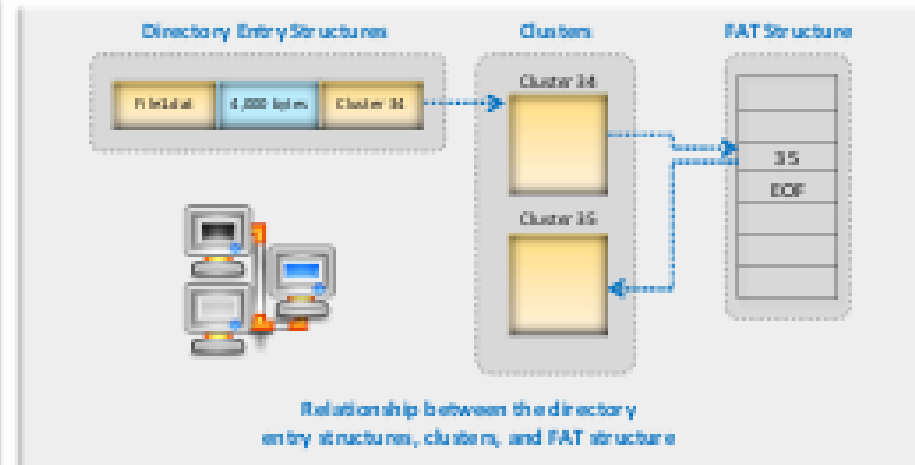


Windows File Systems

File Allocation Table (FAT)

- ➔ The FAT file system is used with DOS, and it was the first file system used with the Windows OS
- ➔ It is named for its method of organization, the file allocation table, which resides at the **beginning of the volume**
- ➔ FAT has three versions (FAT12, FAT16, and FAT32), which differ in terms of the **size of the entries in the FAT structure**

System	Bytes Per Cluster within File Allocation Table	Cluster Limit
FAT12	1.5	Fewer than 4087 clusters
FAT16	2	Between 4,087 and 65,536 clusters, inclusive
FAT32	4	Between 65,536 and 268,435,456 clusters, inclusive



New Technology File System (NTFS)



NTFS is the **standard file system** of Windows NT and its descendants Windows XP, Vista, 7, 8.1, 10, Server 2003, Server 2008, Server 2012, Server 2016 and Server 2019



From Windows NT 3.1 onwards, it is the default file system of the Windows NT family



It has several improvements over FAT such as improved **support for metadata** and the use of **advanced data structures** to improve performance, reliability, and disk-space utilization, as well as additional extensions such as security access-control lists and file system journaling



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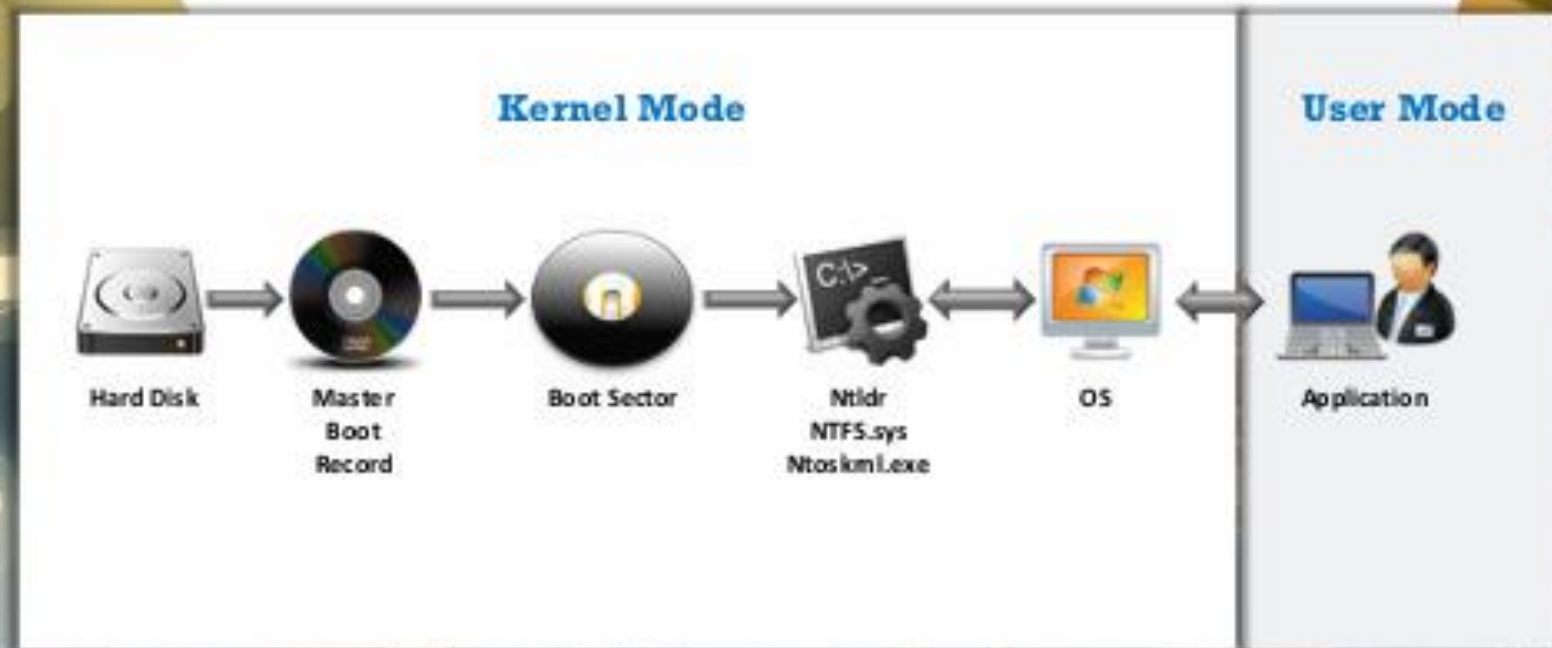
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NTFS Architecture



Encrypting File Systems (EFS)

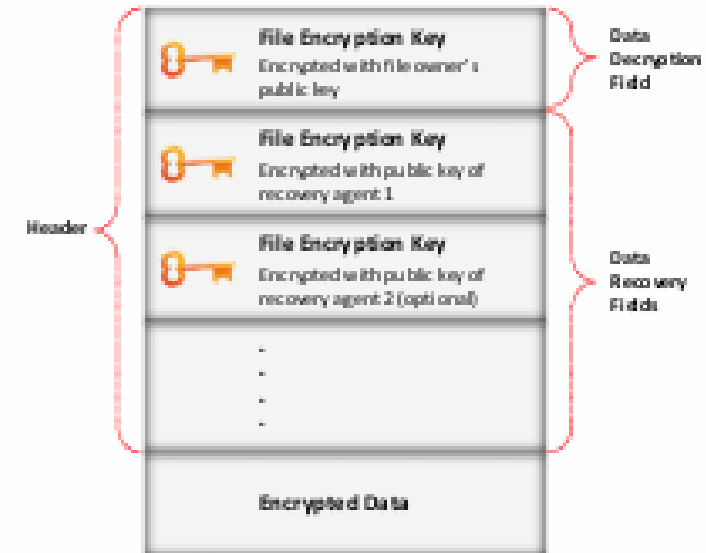
➡ Encrypting File System (EFS) was first introduced in version 3.0 of NTFS and offers file system-level encryption

➡ This encryption technology maintains a **level of transparency** to the user who encrypted the file, which means there is no need for users to decrypt the file to access it to make changes

➡ After a user is done with the file, the **encryption policy** is automatically restored

➡ When any unauthorized user tries to access an **encrypted file**, they are **denied access**

➡ To enable the encryption and decryption facilities, a user must set the **encryption attributes** of the files and folders they wish to encrypt or decrypt



Linux File Systems



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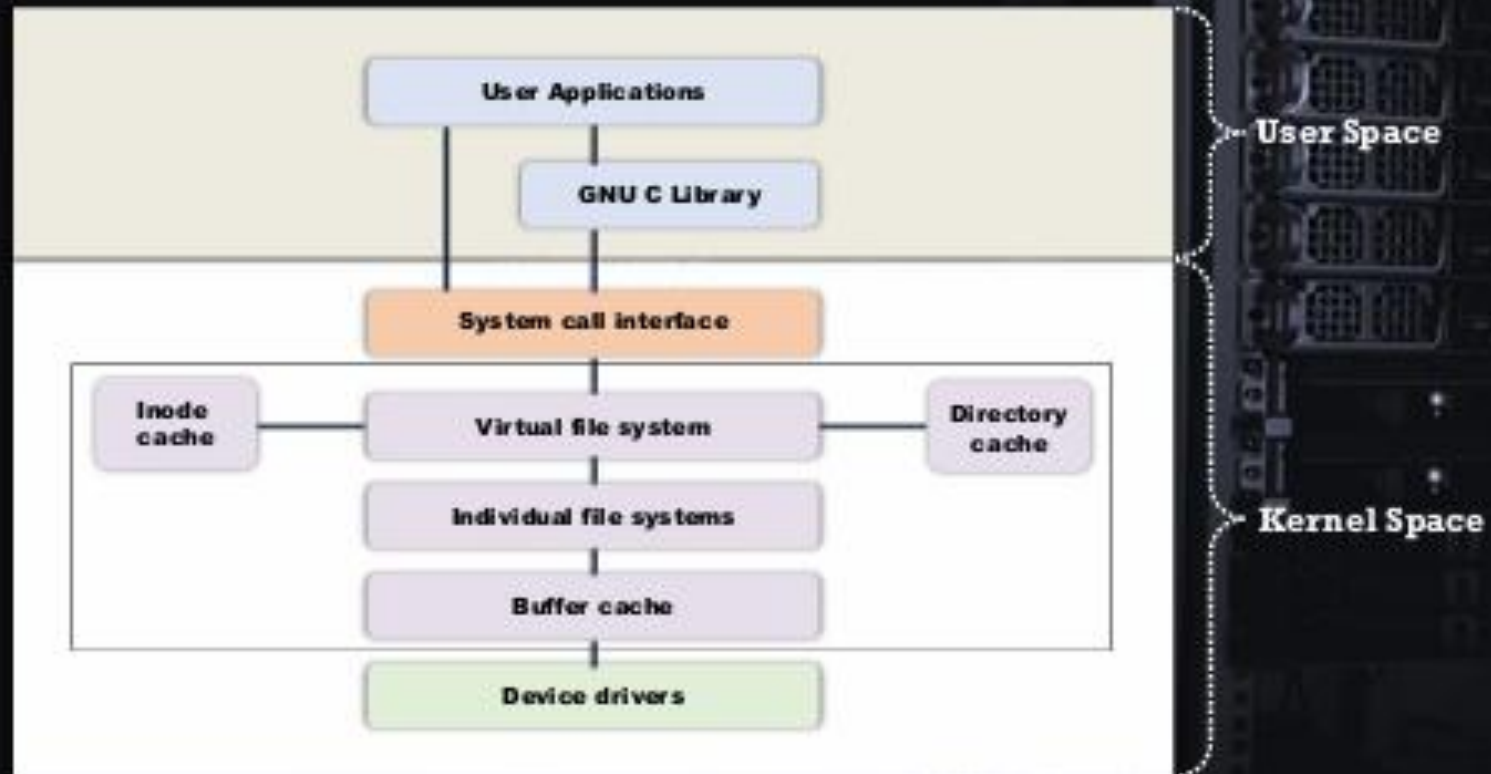
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Linux File System Architecture



Filesystem Hierarchy Standard (FHS)

Table displaying directories and their description specific to the FHS

Directory	Description
/bin	Essential command binaries; e.g., cat, ls, cp
/boot	Static files of the boot loader; e.g., kernel, initrd
/dev	Essential device files; e.g., /dev/null
/etc	Host-specific system configuration files
/home	Users' home directories, which hold saved files, personal settings, etc.
/lib	Essential libraries for the binaries in /bin/ and /sbin/
/media	Mount points for removable media
/mnt	Temporarily mounted file systems
/opt	Add-on applications or software packages
/root	Home directory for the root user
/proc	Virtual file system providing process and kernel information as files
/run	Information about running processes; e.g., running daemons, currently logged-in users
/sbin	Contains the binary files required for working
/srv	Site-specific data for services provided by the system
/tmp	Temporary files
/usr	Secondary hierarchy for read-only user data
/var	Variable data; e.g., logs, spool files, etc.



The **Filesystem Hierarchy Standard (FHS)** defines the directory structure and its contents in Linux and Unix-like OSes



In the **FHS**, all files and directories are present under the root directory (represented by /)



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Extended File System (ext)

The extended file system (ext) is the first file system for the Linux OS to overcome certain limitations of the **Minix file system**

1

2

It has a maximum partition size of 2 GB and a maximum filename size of 255 characters

3

It removes the two major Minix file system limitations: a **maximum partition size of 64 MB** and **short filenames**

4

The major limitation of this file system is that it does not support separate access, inode modification, and data-modification timestamps

5

It was replaced by the **second extended file system** (ext 2)

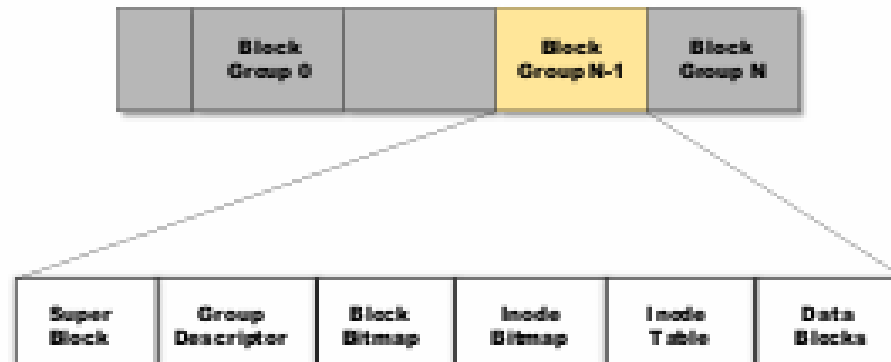


Second Extended File System (ext2)



- ❑ ext2 is a standard file system that uses improved algorithms compared to ext, which greatly **enhances its speed**; further, it maintains additional time stamps
- ❑ It maintains a special field in the superblock that keeps track of the file system status and identifies it as either clean or dirty
- ❑ Its major shortcomings are the **risk of file system corruption** when writing to ext2, and the lack of journaling

Physical layout of the ext2 file system:



Third Extended File System (ext3)



ext3 is a journaling version of the ext2 file system and is greatly used in the Linux OS

It uses **file system maintenance utilities** (such as fsck) for maintenance and repair, as in the ext2 file system



It is an enhanced version of the **ext2** file system

The following command converts ext2 to ext3 file system:

```
# /sbin/tune2fs -j <partition-name>
```



ext3 Features



Data Integrity



Speed



Easy Transition



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Journaling File System

-  1. Journaling file systems ensure **data integrity** on a computer
-  2. These file systems consist of a journal that records all the information on the updates that are ready to be applied to the file system before they are applied. This mechanism is referred to as **journaling**.
-  3. Journaling prevents **data corruption** by restoring the data on the hard disk to the state it existed in before the occurrence of a system crash or power failure. This helps the system to resume the completion of tasks or updates that were interrupted by an unexpected event.
-  4. ext3, ext4, ZFS, and XFS are some of the **examples** of journaling file systems in Linux. Because of its stability, ext4 is the most commonly implemented file system on Linux systems.



Fourth Extended File System (ext4)



- ❑ ext4 is a journaling file system developed as the **replacement** of the commonly used **ext3 file system**
- ❑ With the incorporation of new features, ext4 has significant **advantages** over **ext3** and **ext2** file systems, particularly in terms of performance, scalability, and reliability
- ❑ It supports Linux Kernel v2.6.19 onwards

Key Features





macOS File Systems



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macOS File Systems

UNIX File System (UFS)

- ❖ UFS is derived from the **Berkeley Fast File System (FFS)** that was originally developed at Bell Laboratories from the first version of UNIX FS
- ❖ All BSD UNIX derivatives including FreeBSD, NetBSD, OpenBSD, NeXTStep, and Solaris use a variant of UFS

Hierarchical File System (HFS)

- ❖ Developed by **Apple Computer, Inc.** to support macOS

HFS Plus

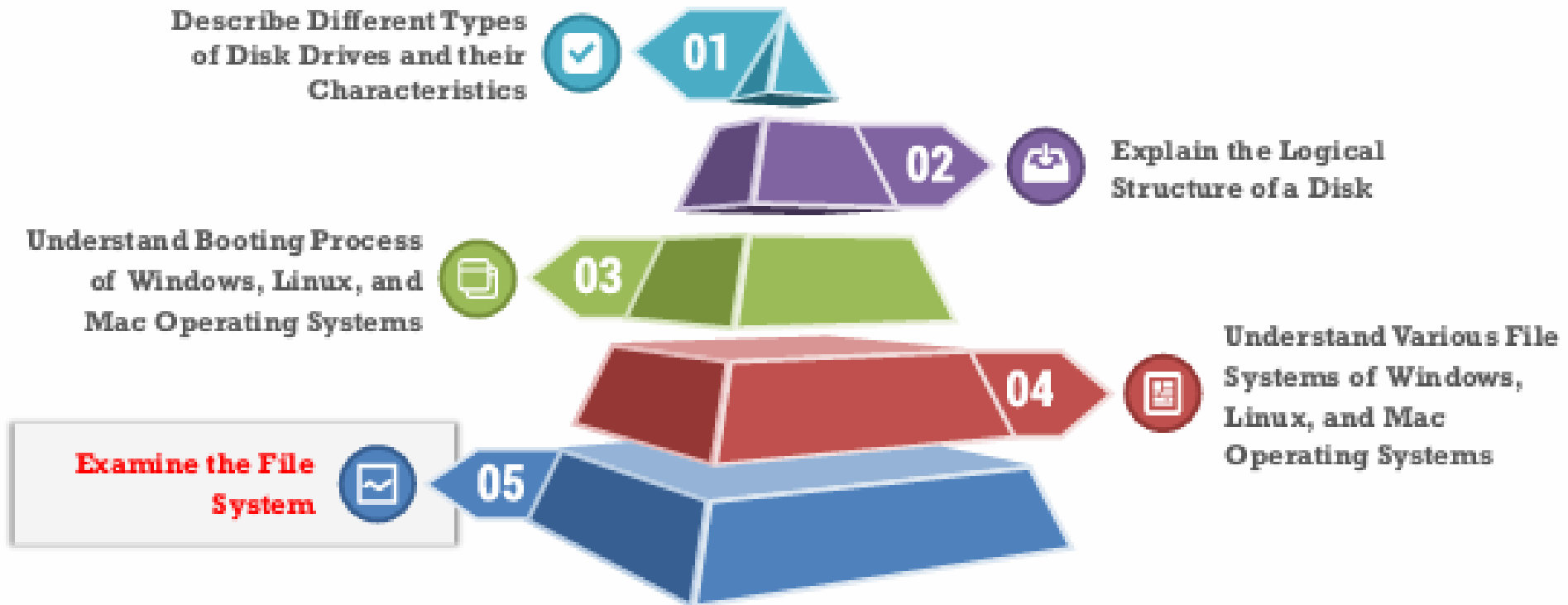
- ❖ HFS Plus (HFS+) is the successor to HFS and is used as the **primary file system** in Macintosh

Apple File System (APFS)

- ❖ It is a proprietary file system developed by Apple Inc. for macOS 10.13 and later versions

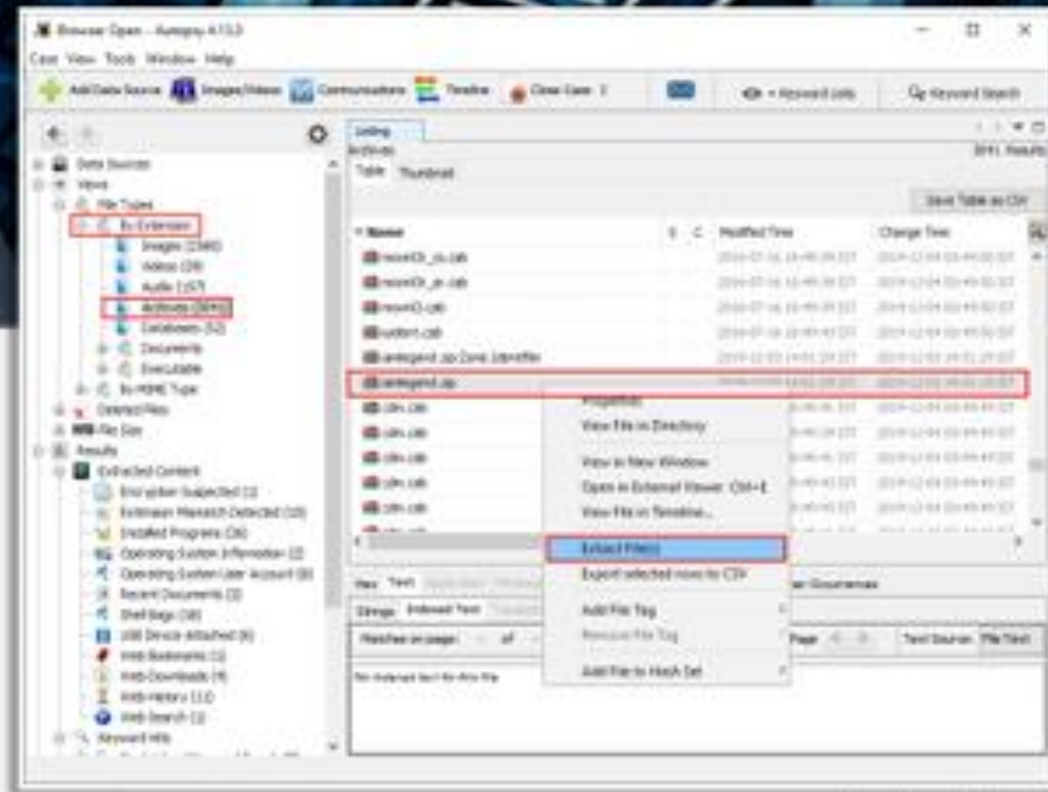


Module Flow



File System Analysis Using Autopsy

- ❑ Autopsy is a digital forensics platform and **graphical interface to The Sleuth Kit (TSK)** and other digital forensics tools
- ❑ It can be used to investigate activities on a computer



<https://www.sleuthkit.org>



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File System Analysis Using The Sleuth Kit (TSK)

- 01 The Sleuth Kit (TSK) is a library and a collection of command-line tools that allow the **investigation of volume and file system data**
- 02 The file system tools allow you to examine file systems of a suspect computer in a non-intrusive fashion
- 03 It supports DOS partitions, BSD partitions (disk labels), Mac partitions, Sun slices (Volume Table of Contents), and GPT disks
- 04 It analyzes **raw (i.e. dd)**, **Expert Witness (i.e. EnCase)**, and **AFF** file systems and disk images
- 05 It supports the NTFS, FAT, ExFAT, UFS 1, UFS 2, ext2, ext3, ext4, HFS, ISO 9660, and YAFFS2 file systems

Note: To perform analysis, create a forensics image **.dd or .E01** of a hard disk or pen drive using disk imaging tools. Here, we have created a forensics image of a pen drive in the .E01 format using AccessData FTK Imager.



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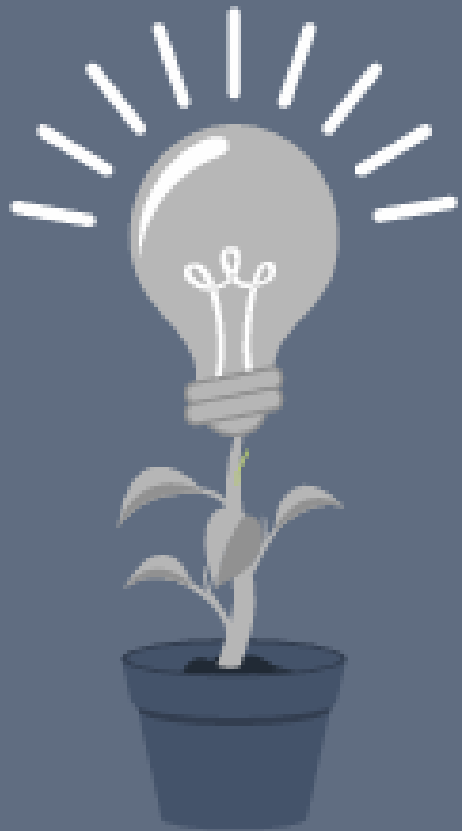
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Module Summary



This module has discussed the different types of storage drives and their characteristics



It has discussed the logical structure of a disk



It has also discussed the booting process of Windows, Linux, and Mac operating systems



This module has also discussed the various file systems of Windows, Linux, and Mac operating systems



Finally, this module ended with a detailed discussion on examining file systems using Autopsy and The Sleuth Kit



In the next module, we will discuss in detail on data acquisition and duplication

REFERENCES

- **Digital Forensics Essentials – 2021 by EC-Council**